

Predicting Abalone Age: A Comparative Study Between a Statistical and an Algorithmic Approach

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Abstract—This work aims to provide a comparative analysis of the best-fit model for Abalone age prediction between statistical and algorithmic models. Research has been conducted to determine the two best-performing models from each category, accompanied by a comparative analysis of the most optimal model between our chosen XGBoost and Polynomial Regression models.

The dataset provided is part of the “Data Mining” course material. In compliance with the guidelines, the data were pre-processed (including missing values removal, encoding, cleaning, and normalization). Feature selection and engineering were applied jointly and separately for both models to maximize performance.

Both models show strong results; however, the algorithmic model demonstrates better predictive power due to its ability to learn complex non-linear patterns. XGBoost performs better thanks to its boosted decision-tree structure and optimized hyperparameters.

I. INTRODUCTION

Abalone species are molluscs that belong to the Haliotidae family. Their biological anatomy consists of a moderately to heavily calcified, snail-like flat shell. In total, there are 56 valid species with an additional 18 subspecies. Their shells contain visible rings and protect the internal soft-tissue components of the organism [1].

The age of abalone is calculated by a process of cutting the shell through the cone, staining it, and counting the number of rings under a microscope [2]. This is generally a quite expensive and time-consuming method to determine the age of abalone.

Several predictive modelling methods can be used to estimate abalone age solely from morphological measurements. Motivated by [3] and the debate among modelling cultures, this work introduces two models: one that better suits statistical modelling (such as Polynomial Regression) and another better suiting algorithmic modelling (such as XGBoost). Both excel at modelling linear and nonlinear relationships respectively, which makes them suitable for determining the nature of the features and the behaviour of changes in age relative to morphological characteristics.

All stages of the modelling pipeline have been examined, from pre-processing, data partitioning, and feature analysis up to model evaluation. Every implementation step and methodological decision for these models is discussed thoroughly, and the reasoning and effectiveness behind each choice are evaluated.

The paper is organized as follows: Section II describes the dataset and the full pre-processing pipeline, while Section III presents the feature selection, engineering procedures, and

data-splitting strategy used for both models. Section IV details the modelling approaches and validation strategies, Section V reports and compares the results, and Section VI concludes the study with final insights and implications.

II. DATA CHARACTERISTICS AND PRE-PROCESSING

The dataset used contains 4000 observations of Abalone species. Each abalone is represented by 9 features stemming from different dimensions, weights, and categorical separations, such as the ‘sex’ attribute. The Number of rings feature is included as well, which is the key variable in determining the age (in years) of abalone and is estimated using the standard biological approximation:

$$\text{Age} = \text{Rings} + 1.5$$

Table I shows the list of features of the original data before any pre-processing or transformation applied at this stage.

TABLE I
DESCRIPTION OF ABALONE DATASET FEATURES

Feature	Description
Sex	Category: M = Male, F = Female, I = Infant
Length	Longest shell measurement (mm), float
Diameter	Shell diameter perpendicular to length (mm), float
Height	Height with meat in shell (mm), float
Whole weight	Weight of whole abalone (g), float
Shucked weight	Weight of body without shell (g), float
Viscera weight	Organ weight (g), float
Shell weight	Weight of shell only (g), float
Number of rings	Integer count of growth rings (1–29)

Wrong and unprepared data blur the ability of the model to recognize and interpret underlying patterns in a dataset. Pre-processing is a key Data Mining process where a series of procedures are executed, resulting in a dataset from which the model is able to learn efficiently. Any action at this stage, whether positive or negative, has the same resulting effect on the final performance of a model.

The pre-processing process consists of: (a) detecting and replacing missing values; (b) feature encoding; and (c) normalization and (d) data splitting[4]. In the following, each of these steps is introduced, describing the main solutions available for each and the reasoning behind choosing the specific techniques used.

A. Replacing Missing Values

Real-world data has the tendency to be unstable in the context of being incomplete and filled with missing values. Some of the main methods for dealing with missing values

are: (a) Ignore/Delete the row where missing value is present; (b) replace missing values with class mean; and (c) replace missing values with the mean of k-nearest neighbours [4].

The process of replacing missing values for this data set was quick and computationally low. The dataset contained only 2 missing values in total and since this number is significantly low, we removed the rows that contained the missing values, from the dataset.

B. Feature Encoding

The majority of machine learning models are unable to function with non-numerical data in the data set. The feature encoding process converts categorical values into numerical data [5]. Some of the main techniques of feature encoding are:

- *Label encoding* converts non-numerical categorical data by assigning a unique integer to each category. For example, Red, Blue, and Green are mapped to 0, 1, and 2 respectively.
- *Ordinal encoding* works only when the categorical values have a hierarchy. It assigns integers based on the level of hierarchy each value belongs to. For example, a feature named 'size' is mapped to 1, 2, and 3 for small, medium, and large respectively.
- *One-hot encoding* splits a categorical feature into binary features, where each new feature corresponds to one category and is assigned a value of 1 if present and 0 otherwise.

Following this definition, one-hot encoding was applied to this data set through a Pandas function, where the only categorical feature, 'sex', was dropped and replaced with three new features representing the observed categories: 'i' (infant), 'm' (male), and 'f' (female).

C. Normalization

Normalization is one of the pre-processing techniques where the data is either rescaled or transformed to allow each feature to have an equal contribution [1]. There are numerous types of normalization, but among the most important are:

- min/max [0,1] normalization, where values are rescaled to a range of 0 to 1;
- min/max [a,b] normalization, where values are rescaled to a range of a to b;
- z-score normalization, where values are standardized relative to their mean and variance.

Our method of choice is the [0,1] min/max normalization, and it was applied in two stages. After the data-splitting stage, normalization was performed to prevent data leakage. The training and test sets were both normalized using the min/max values of the training set, and the same parameters were used for the corresponding test set [6]. Alternatively, if the training set were normalized using the data set's global min/max values, it would result in the training set learning information from outside its own scope.

During model implementation, normalization was applied within each fold of the k-fold validation process. To prevent

data leakage, each fold was normalized using its own training min/max, and the same parameters were then applied to the corresponding validation fold.

D. Data Splitting

The dataset was divided into a training and a testing set with an 80/20 split ratio. The training set was used to train all the models, and it also went through a k-fold cross-validation process for each model. Data leakage was avoided to ensure that the models reached optimal performance. The test set did not go through any training or validation procedures and was kept for use at the end for performance evaluation. This allowed the test results to be an accurate representation of the models' efficiency.

III. FEATURE ANALYSIS

Feature analysis has played a crucial role in shaping the efficiency of our models by ensuring that the right learning conditions are present for optimal performance. Predominantly, both models share the same underlying feature selection and engineering strategy; however, there are slight distinctions between them based on their learning preferences. Even though the engineered features are based on the same biologically motivated and data-driven principles, we find that XGBoost benefits from a more narrowed-down approach compared to its competing model. Polynomial Regression appreciates a larger set of features mainly due to its flexibility; combined with our Lasso regularization implementation, it is able to determine the most valuable features, ultimately aiding final performance.

A. Feature Selection

Looking at studies on abalones, we can conclude that most species do not show a significant difference in age between 'male' and 'female' specimens. These studies also suggest that growth and age development are predominantly connected to environmental factors rather than gender [7], [8]. This, combined with the general biological intuition that gender is not a primary determinant of age in most species, has led us to consider that the 'male' and 'female' feature does not have significant learning weight in predicting abalone age, thus leading to its removal.

To ensure optimal model performance, we have reduced the chance for our model to learn from what we classify as noise. From our previously derived features from the one-hot encoding process, the 'sex' feature has been dropped, leaving a single one-hot encoded feature, *infant*, to be the only representative of 'sex' in the dataset, where an abalone is either an infant or a mature specimen.

Similar to our feature selection procedures, we have guided our feature engineering approach by transforming the data into more meaningful biological relationships [9]. While the raw data captures detailed morphological changes of an abalone, additional features were derived to capture how specific components relate and scale within its anatomy [10]. Ultimately, this allows the models to access more informative patterns that were not previously detectable in the raw features [11].

B. Feature Engineering for XGBoost Model

The feature engineering process for this model started with raw features plus four biologically motivated ratios and compositions: *meat_ratio*, *viscera_ratio*, *shell_ratio*, and *soft_weight*. Upon experimentation, we discovered that the model performs best when only *meat_ratio* and *viscera_ratio* are present, excluding the remaining engineered features. These ratios represent the abalone's shucked and viscera weight relative to its whole weight. XGBoost benefits from this more narrowed-down approach due to its tree-based learning nature [11]. As its methodology is based on randomness, allowing the model to learn from more focused and highly informative features improves overall performance.

C. Feature Engineering for Polynomial Regression Model

In contrast, the Polynomial Regression model kept all the previously stated feature engineered features. Polynomial expansion heavily relies on representing nonlinear trends, so having a wider range of relationships beyond the raw data increases the scope from which the model can learn these patterns. This can often lead to overfitting, so regularization approaches are applied to the model as well [11]. After implementing Lasso regularization, the model ends up retaining 26 terms, including raw features such as *viscera_ratio*, several squared terms, and important interaction terms such as *infant* $\times$ *length* and *shell_weight* $\times$ *soft_weight*. This shows that the model has benefited from an extended selection of features while retaining the ability to remain balanced by keeping only the most important information for its final performance.

Table II below summarizes all engineered features used across both models and those ultimately retained.

TABLE II
ENGINEERED FEATURE USAGE ACROSS MODELS

Model	Engineered Features Used	Final Engineered Terms Kept
XGBoost	<i>meat_ratio</i> , <i>viscera_ratio</i>	<i>meat_ratio</i> , <i>viscera_ratio</i>
Poly. Reg. (Before LASSO)	<i>meat_ratio</i> , <i>viscera_ratio</i> , <i>shell_ratio</i> , <i>soft_weight</i>	All engineered features included before expansion
Poly. Reg. (After LASSO)	Same engineered features; LASSO filters terms	Selected raw features together with LASSO-retained polynomial and interaction terms (e.g., <i>viscera_ratio</i> , squared terms, key interactions)

IV. MODELS ANALYSIS

A. Polynomial Regression with Lasso Regularization

1) *Model Selection and Justification*: Polynomial Regression was the method of choice to best represent the statistical approach inspired from Breiman's framework [3]. Contrary to linear regression, polynomial regression is able to also capture non-linear morphological patterns [12], [13], [14]. Through quadratic terms and feature interactions our model can gain insight from both linear and non-linear patterns contributing to optimal performance results [9], [15].

Among non-linear statistical methods (such as splines, GAMs, and kernel methods), polynomial regression was chosen with the purpose of having a model that can be computationally efficient, involve a transparent feature process and maintain model interpretability [16], [17], [18]. Lasso regularization and degree-2 expansion was chosen over higher-degree polynomials so that we maintain performance and prevent overfitting [9].

Among Ridge (L2), ElasticNet and Lasso (L1) regularization [19], [9], the latter was chosen as best fit for our preferred outcome. The degree-2 polynomial expansion generated 90 terms from its original 12 features. Ridge regularization option was excluded due to not being able to remove coefficients completely but rather just shrink them [9], [20]. Combination of L1 and L2 would not be also feasible because it introduces an additional mixing parameter that requires additional tuning with no clear benefit for this task [21]. Lasso's explicit method drives our unnecessary coefficients to exactly zero, allowing room for efficiency and knowing exactly what features and combinations are important to us [19], [22].

Through cross-validated alpha tuning, LassoCV with 10-fold CV, retained 26 terms including raw features (*infant*, *length*, *diameter*), engineered ratios (*viscera_ratio*), squared terms (*length*², *diameter*²), and key interactions (*infant* $\times$ *length*, *shell_weight* $\times$ *soft_weight*). 71% of polynomial terms were eliminated, ensuring predictive power and interpretability while reducing complexity [9], [23]. The complete list of 26 retained terms is provided in Table III.

To ensure that we have resulted in a good Alpha value, a nested K-fold cross-validation procedure was used to train the model on its validation stage and inside it derive a separate Alpha value to compare. [24], [25].

2) *Model Training*: Succeeding the pre-processing and feature-related procedures described in the previous sections, our degree-2 polynomial regression model was created using the normalized training data and generating the required feature space through the *PolynomialFeatures* function. This transformation enlarged the feature space to 90 dimensions, giving the model the opportunity to learn from a more expanded set of data. To ensure that the model did not learn anything external, normalization was applied to the training and test data using only the parameters of the training set.

Polynomial regression models benefit from a large set of terms and features, but they also risk learning the data too well and overfitting. To avoid this, the next stage was identifying which set of features was most valuable and informative. Our final model was trained using a reduced set of features generated through a 10-fold internally cross-validated LassoCV applied to the training set and then validated and tested on the test set. LassoCV evaluates a series of strengths (alphas) for each fold and selects the one that results in the lowest cross-validated error [23], [26]. This ensures the chosen value reflects the best method to prevent overfitting, leading to less complex models and coefficients of unimportant terms being shrunk to exactly zero. The final model ends up learning from the most important polynomial features while noise is removed

from the model.

Using the optimal alpha and the selected reduced feature set, the final model was trained on the full training data and subsequently used to generate predictions on the held-out test set, ensuring that the reported performance reflects behaviour on data the model has never seen before.

3) *Model Validation*: After model training, the next step was to evaluate how well our model performs on unseen data. Even with our optimally derived alpha coefficient, we still need to ensure that the model is not simply memorizing patterns from the training set but is capable of generalizing to any data it encounters. To guarantee this, a nested K-fold cross-validation strategy was implemented [24], [25]. This approach separates training from evaluation by assessing the performance of the obtained parameters on new data.

Our nested K-fold structure consists of a loop with 10 outer folds, where in each fold a portion of the data was held out as test data and the remaining data was used to train the model. Inside each outer fold, an inner 5-fold LassoCV was implemented to obtain the best alpha value. Similar to our final alpha coefficient, this loop compares several regularization strengths and determines the one with the lowest cross-validated error on the training portion of that fold [26]. The result is that we can compare the final alpha coefficient to how consistent and reliable it is throughout the validation process.

To keep the validation process fair, all data preprocessing steps were applied within each fold. Normalization and polynomial feature expansion were fitted only to the training portion and then applied to the validation portion. This method ensures that the model never sees any information outside its training set. By preventing data leakage, we obtain more realistic estimates of how the model will behave in practice.

B. XGBoost Regression

1) *Model Selection and Justification*: In contrast to our statistical approach, we also explored finding the best algorithmic model for our problem. The goal was to use a method that could capture the nonlinear patterns in our dataset more efficiently. Although models such as neural networks, decision trees, random forests, and gradient boosting methods are capable of this, our decision was guided mainly by the characteristics of the abalone dataset [27], [28], [29], [30]. Since the dataset is moderate in size and consists mostly of numerical tabular features, decision-tree-based models stood out as the most suitable option. Prior research has shown that these models learn complex patterns well, require little preprocessing, and remain computationally efficient on tabular datasets [31], [32].

Single decision trees were not selected because they can overfit easily and are not stable on their own. Random forests address this by averaging multiple trees, but they do not learn sequential improvements and may miss subtler patterns in the data. This led us to consider gradient boosting methods, which build trees sequentially, with each tree attempting to correct errors from the previous ones. Among these boosting algorithms, XGBoost was chosen instead of alternatives such

as LightGBM and CatBoost [33], [31]. CatBoost performs best on datasets with many categorical features, which is not the case here, while LightGBM can be sensitive on smaller datasets and sometimes requires more careful tuning. XGBoost provides a strong balance of regularization, stability, and performance, with built-in mechanisms such as shrinkage, subsampling, and depth control to prevent overfitting [34][27], [28], [29], [30].

These combined strengths made XGBoost the most reliable algorithmic model for our prediction task. As with our statistical model, cross-validation was used to ensure that the chosen hyperparameters generalize well and that the final performance estimate is unbiased.

2) *Model Training*: Having completed the pre-processing techniques outlined in the prior sections, such as one-hot encoding of the sex feature and the generation of the engineered variables *meat_ratio* and *viscera_ratio*, the XGBoost regressor was trained on the enhanced dataset. XGBoost is a tree-based model, and polynomial feature expansion is not involved in its process; instead, it captures nonlinear patterns and interactions more naturally [34]. Although XGBoost does not require feature scaling, normalization was applied consistently across models for comparability.

To increase performance, hyperparameter tuning was carried out on the model. Due to its efficient optimization of complex search spaces, Hyperopt was chosen as the preferred method for this task. Based on Tree-structured Parzen Estimators (TPE), Hyperopt assesses the performance of past trials and concentrates its search on parameters that perform better [35]. This method is more effective than traditional techniques such as GridSearch or RandomizedSearch, primarily because these methods are not capable of learning from previous evaluations. Hyperopt learns from past outcomes by using a Bayesian optimization method, which develops a probabilistic model of the search space based on prior trials [36]. As tuning continues, the model updates by focusing on parameters that have shown better performance while steering away from areas that produced poor results. This strategy has resulted in a significant performance boost compared to initial stages without tuning.

To determine the search space for Hyperopt, the main XGBoost hyperparameters were used, including the number of estimators, learning rate, maximum tree depth, subsampling ratio, column sampling ratio, minimum child weight, and gamma. These ranges were selected based on the recommendations of the official XGBoost documentation [37] and established practices in the literature [34]. Sixty parameter combinations were processed by Hyperopt, and each trial was evaluated using a 10-fold cross-validation loop on the training data. As Hyperopt tested multiple combinations across all folds, the one achieving the strongest average R^2 was chosen as the best-performing and most generalizable configuration.

Upon selecting the best hyperparameters, the final XGBoost model was trained on the full training set. Predictions were computed on the held-out test set, generating the final MAE, RMSE, and R^2 metrics. To interpret model behaviour, SHAP

values were computed, enabling the analysis of feature impact on the final prediction [38].

3) *Model Validation*: Upon successful derivation of the best hyperparameters through Hyperopt, the next step for our XGBoost model was to assess how well it performs on unseen data. Incorporating 10-fold cross-validation into the tuning process ensured that each hyperparameter set was evaluated across multiple splits of the training data rather than on a single partition [39], [40]. By testing the parameters repeatedly on randomly selected subsets, the tuning process becomes more reliable and helps identify configurations that consistently perform well. This process not only reduces the risk of overfitting but also improves the model's predictive performance.

Once the optimal hyperparameter was set, the model was trained again on the entire training set. The held-out test set, which we kept separate from the beginning and ensured no data leaked into, was then used to test our final XGBoost model's performance on unseen data [41], [42]. This separation was key to ensuring that the reported performance reflects the true readiness of the model for new data, delivering an unbiased estimate of real-world predictive capabilities.

To gain a more detailed understanding of our model's performance, SHAP values were computed on the training set, allowing us to observe the contribution of each feature to the final prediction. These values offer a transparent interpretation of how the model utilizes morphological measurements to estimate the age of abalone [38]. Rather than evaluating the model solely on prediction accuracy, we also investigated the mechanisms behind its decisions, highlighting the patterns and relationships that XGBoost identified within the data.

V. RESULTS

A. Polynomial Regression

Using a degree-2 expansion and LASSO regularization, the Polynomial Regression model reduced the initial 90 polynomial terms to a sparse and comprehensible set of 26 predictors. The preserved items included key raw morphological features (length, diameter, height), nonlinear effects (length², diameter²), and meaningful associations involving the infant feature and engineered variables such as *viscera_ratio*. Essentially, both linear and nonlinear relationships contribute to age estimation, while LASSO successfully removed raw or engineered features that are either noise or not relevant to the goal.

Performance results on the test set showed that our model achieved an MAE of 0.0552 and an RMSE of 0.0779 based on the scaled values. Converted back to the original scale, this means that the model typically predicts 1.54 to 2.2 years off from the real abalone age. The R² score of 0.6081 indicates that the algorithm explains approximately 60.7% of the variability in age. We conclude that the model detects the main structural patterns in the data while indicating that the remaining variation is due to biological noise or nonlinear effects that cannot be captured by a second-degree polynomial.

As a statistical model, Polynomial Regression provides insight through its error metrics. MAE represents the average prediction error, and the deviations observed are generally acceptable. RMSE, which penalizes larger errors more heavily, suggests that the model tends to make larger mistakes on older abalones, where morphological variation is higher. The small distance between the two error values suggests that large deviations are rare. Overall, the metrics indicate that the model is stable, reasonably accurate, and produces consistent predictions across most of the dataset.

Our diagnostic plots offer additional support for these claims. With a slight underestimation at older ages, the predicted-vs-actual plot exhibits a strong linear trend. The Q-Q plot shows modest deviations in the upper quantiles, reflecting occasional larger errors for older specimens, whereas residual plots show mostly random error distributions with slight heteroscedasticity. Taken together, the Polynomial Regression model strikes a balance between accuracy and transparency by producing predictions that are both interpretable and biologically meaningful.

The diagnostic plots used to evaluate the model's behaviour (Figures 1–4) are provided in the Appendix.

B. XGBoost Regression

The final XGBoost model was trained using the derived hyperparameters on the full training set and assessed on the held-out test set. On the normalized scale, the model produced an MAE of 0.0550, an RMSE of 0.0772, and an R² value of 0.6151. In original units, this translates to the model being off by approximately 1.50–1.60 years. This means that for each age prediction, the model is typically off by one or two years. Compared to our first model, the slightly lower MAE and higher R² indicate that XGBoost detects and explains a larger portion of the nonlinear relationships in the data.

A Predicted vs Actual scatter plot was constructed to visualize how closely the model's predictions align with true abalone ages. The model is capable of producing accurate predictions across most ages, but a few points fall farther from the perfect-fit line, indicating that errors are larger for older abalones. This pattern aligns with the error metrics: the low MAE reflects relatively small errors across predictions, while the slightly higher RMSE captures increased error at higher ages. With an R² of 0.6151, the model captures over 61.5% of the variation in abalone age, with the remaining unexplained variance likely due to biological variability and noise rather than model limitations.

To understand the model's behaviour, SHAP values were generated as an interpretability method. Bar plots and the SHAP summary show that *shell_weight* is the most significant predictor, consistently increasing the expected age as shell mass increases. This trend aligns with biological intuition, as many species become heavier with age. The second most influential variable was the engineered feature *meat_ratio*, indicating that the model relies on soft-tissue composition when estimating age. Features such as height, *viscera_ratio*, and length also played notable roles, capturing geometric

and proportional aspects of growth. In contrast, raw weight measures such as *whole_weight* and *shucked_weight* had lower SHAP values, suggesting that XGBoost depends more on derived structural and ratio-based descriptors than absolute mass.

In summary, the XGBoost model offers strong predictive performance and provides meaningful interpretability through the generated plots and SHAP analysis. The model reliably captures complex nonlinear and engineered relationships between morphological features, resulting in accurate and interpretable predictions. XGBoost remains flexible and capable of modelling intricate patterns and interactions within the data, making it well suited for our goal. The diagnostic plots used to evaluate the model's behaviour (Figures 5–7) are provided in the Appendix.

C. Critical Comparison

1) *Performance Summary*: All models displayed strong predictive capacities, but XGBoost was able to achieve slightly greater accuracy and generalization. On the test set, XGBoost produced an MAE of 0.0550 (≈ 1.50 years), an RMSE of 0.0772 (≈ 2.16 years), and an R^2 of 0.6151, compared to Polynomial Regression's MAE of 0.0552, RMSE of 0.0779, and R^2 of 0.6081. Although the differences are not substantial, XGBoost shows that it can provide more reliable performance and is more flexible in adapting to the structure of the dataset. The generated Predicted-vs-Actual plots also support this claim. Both models perform similarly, as each succeeds in producing small errors for younger to medium-aged abalones but struggles more with older specimens. However, when comparing their overall performance, XGBoost maintains a tighter clustering of predictions, which leads to lower error rates. Considering the quadratic nature of Polynomial Regression, its slightly weaker performance can be attributed to its struggle in describing complex, irregular patterns in the data.

TABLE III
PERFORMANCE COMPARISON BETWEEN MODELS

Model	MAE	RMSE	R^2
Poly. Reg. / XGBoost	0.0552 / 0.0550	0.0779 / 0.0772	0.6081 / 0.6151

2) *Interpretability vs. Performance*: The main difference, although not substantial, between the models is found in their interpretability. Polynomial Regression is based on a clear mathematical structure, and with LASSO regularization handling complexity and overfitting, only 26 terms are retained. Each term can be assessed directly, making the model easy to interpret and fit for tasks where the impact of individual features must be understood. XGBoost, in contrast, is more complicated internally due to its ensemble structure. However, using SHAP values meaningful insights into how features influence predictions were clearly provided. This allows interpretability even though the model does not have an inherently simple structure that is easy to understand. Thus, the Polynomial model offers clearer inherent interpretability, whereas XGBoost trades some transparency for higher predictive strength.

3) *Practical Recommendations*: Down the line, the choice between the two models depends on the objective of the task at hand. If obtaining more accurate predictions, such as in ecological monitoring or fisheries management, is the goal, XGBoost is the more practical option, as it adapts quickly to nonlinear patterns with less manual effort in model development. Although XGBoost is less transparent by design, SHAP explanations narrow the interpretability gap enough for most applied tasks. However, Polynomial Regression offers advantages in settings where the aim is to understand biological relationships rather than simply predict age, allowing researchers to use its explicit equation to see how each morphological trait contributes to the final estimate, making it useful for multiple applications in that field. In practice, Polynomial Regression functions as a clear and biologically interpretable baseline, while XGBoost serves as the more robust and dependable choice for real-world prediction where performance and generalization are the priority.

VI. CONCLUSION

This study compared a statistical modelling approach—Polynomial Regression with LASSO regularization—and an algorithmic model, XGBoost, to identify the most suitable method for predicting abalone age from morphological measurements. Both models achieved strong and consistent performance, with XGBoost showing a small but steady improvement across all evaluation metrics. These differences are modest, which indicates that Polynomial Regression remains a capable and well-behaved option, even though XGBoost generalizes slightly better overall.

Interpretability was also considered in the comparison. Polynomial Regression provides a clear mathematical equation that shows how each morphological feature contributes to the prediction, making it especially useful when the goal is to understand growth patterns or communicate biological relationships. Although XGBoost is less transparent in its structure, SHAP explanations reduce this gap by offering intuitive insights into feature importance. As a result, both models support interpretability, but Polynomial Regression offers a more direct way to study physiological relationships, whereas XGBoost explains its decisions through post-hoc analysis.

When the aim is accurate and reliable age prediction—such as in fisheries management, ecological monitoring, or automated analysis—XGBoost is the more practical choice, as it handles nonlinear patterns automatically, includes built-in regularization, and requires less manual tuning to avoid overfitting. However, if the focus is on understanding how abalone morphology relates to age or on conducting biologically driven analysis, Polynomial Regression may be the preferred model due to its simple and explicit structure. Overall, both models contribute valuable insights, but XGBoost offers the more robust solution for predictive applications, while Polynomial Regression remains the stronger option for interpreting biological relationships in the data.

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APPENDIX

APPENDIX B: POLYNOMIAL REGRESSION SUMMARY

TABLE IV
POLYNOMIAL REGRESSION MODEL SUMMARY AND RETAINED FEATURES

Description	Value
Total Polynomial Features	90
Chosen α (LASSO)	1.3041×10^{-5}
Final Test Performance (Normalized)	
MAE	0.0552
RMSE	0.0779
R ²	0.6081
Final Test Performance (Original Scale)	
MAE (years)	1.5480
RMSE (years)	2.1834
R ²	0.6081
Retained Polynomial Terms (26 Total)	
infant	infant $\times$ height
length	infant $\times$ viscera_weight
diameter	infant $\times$ shell_weight
height	infant $\times$ viscera_ratio
whole_weight	length ²
shucked_weight	length $\times$ viscera_ratio
shell_weight	diameter ²
viscera_ratio	height ²
infant ²	whole_weight ²
infant $\times$ length	shucked_weight ²
infant $\times$ diameter	shucked_weight $\times$ shell_weight
meat_ratio ²	shell_weight $\times$ soft_weight
viscera_ratio ²	shell_ratio ²

APPENDIX C: XGBOOST MODEL SUMMARY

TABLE V
XGBOOST REGRESSION: CROSS-VALIDATION AND TEST PERFORMANCE

Description	Value
n_estimators	455
learning_rate	0.0944
max_depth	3
subsample	0.8097
colsample_bytree	0.9873
min_child_weight	9
gamma	0.0318
MAE (Normalised)	0.05504
RMSE (Normalised)	0.07728
R ² (Normalised)	0.6151
MAE (original)	1.5412
RMSE (original)	2.1639
R ² (original)	0.6151

APPENDIX D: POLYNOMIAL REGRESSION — PREDICTED VS ACTUAL

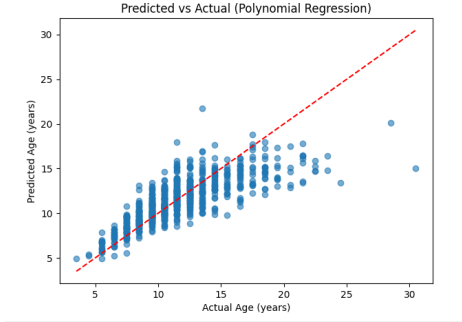


Fig. 1. Polynomial Regression: Predicted vs Actual Plot

In Figure 1 the plot compares the model's predicted ages to the real ages. Each point represents one abalone in the test set. If the model were perfect, all points would lie exactly on the diagonal line. In this model, most points stay close to the diagonal...

APPENDIX E: POLYNOMIAL REGRESSION — RESIDUALS VS FITTED

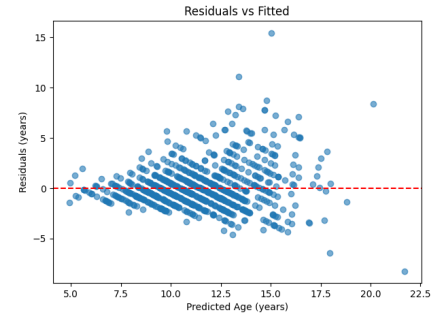


Fig. 2. Polynomial Regression: Residuals vs Fitted Plot

In Figure 2 the plot shows the errors the model makes (residuals) against its own predictions. A good model should not show any clear pattern here. The points should be scattered randomly around the zero line. In our case, the residuals look fairly evenly spread without a strong curve or pattern. This means the model does not consistently overpredict or underpredict across different age ranges. This randomness suggests that the polynomial model is capturing the main relationships correctly and is not missing any obvious trend.

APPENDIX F: POLYNOMIAL REGRESSION — RESIDUAL DISTRIBUTION

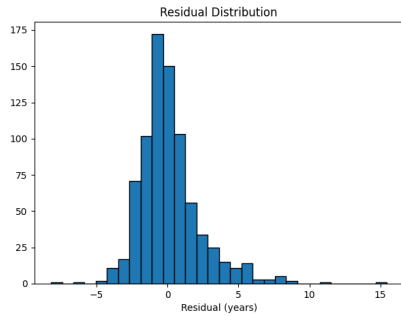


Fig. 3. Polynomial Regression: Residual Distribution

In Figure 3 the histogram shows how the model's prediction errors are distributed. Most residuals cluster close to zero, meaning most predictions are accurate. There are fewer large positive or negative errors, which shows that the model rarely makes very big mistakes. A roughly bell-shaped distribution like this is what we expect from a stable regression model. This confirms that the model generalizes well and does not suffer from major bias toward overprediction or underprediction.

APPENDIX H: XGBOOST — SHAP BEESWARM PLOT

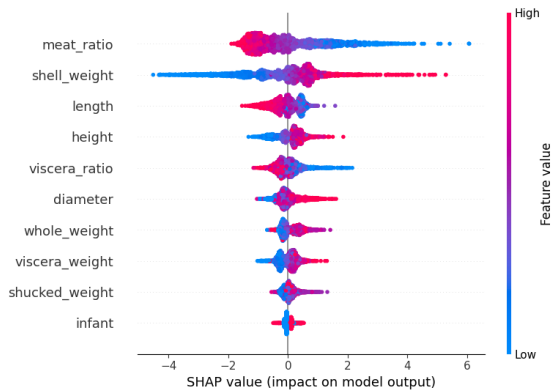


Fig. 4. XGBoost Regression: SHAP Beeswarm Plot

As shown in Figure 4, the SHAP beeswarm plot shows how each feature influences XGBoost's predictions. Each point represents a single sample, and the color shows whether the feature value is high (red) or low (blue). Features such as *meat_ratio* and *shell_weight* have the strongest impact, which matches biological expectations — heavier and meatier abalones tend to be older. High SHAP variation indicates the model is leveraging complex non-linear relationships, contributing to its high predictive accuracy.

APPENDIX I: XGBOOST — SHAP FEATURE IMPORTANCE

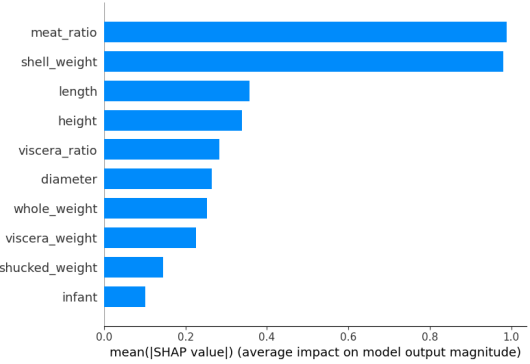


Fig. 5. XGBoost Regression: SHAP Feature Importance

Figure 5 shows the average importance of each feature according to SHAP values. Here, the size of the bar reflects how much each feature contributes to the model's predictions on average. *meat_ratio* and *shell_weight* stand out as the most influential predictors, confirming they contain strong signals about abalone age. Lower-ranked features like *infant* still contribute, but with much smaller impact. This ranking reinforces the stability and interpretability of the XGBoost model.

APPENDIX J: XGBOOST MODEL — PREDICTED VS ACTUAL

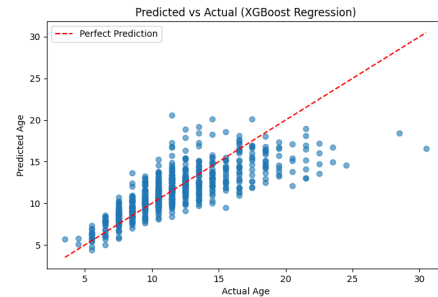


Fig. 6. XGBoost Regression: Predicted vs Actual Plot

In Figure 6, the predicted ages generated by the XGBoost model are compared against the true abalone ages in the test set. Each point represents a single abalone observation. The dashed diagonal line corresponds to perfect prediction, where predicted and actual ages are equal. Most predictions cluster closely around this reference line, indicating strong agreement between predicted and true values across a wide age range. A slight increase in dispersion is visible for higher ages, suggesting that prediction errors tend to grow for older abalones. Overall, the plot confirms that the XGBoost model captures the underlying age trend effectively while maintaining consistent predictive accuracy.